WEEK 3 · LESSON 15 OF 24 · HANDOUT

Neural Networks Explained

Study time: **about 50 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=aircAruvnKk — open the lesson online to watch inside the course.

🎓 **Go deeper:** Harvard CS50 AI, Lecture 5: Neural Networks — https://www.youtube.com/watch?v=mFZazxxCKbw

🎓 **Go deeper:** MIT 6.S191: Introduction to Deep Learning (2024) — https://www.youtube.com/watch?v=ErnWZxJovaM

Your goal in this lesson: Describe the parts of a neural network — neurons, weights, biases, layers and activation functions — and explain why depth helps with images, sound and language.

Inspired by the brain, built with maths

A **neural network** is a model made of many small calculating units called **neurons**, connected in **layers**. It was loosely inspired by brain cells, but it is really a large mathematical function. Neural networks are the engine of **deep learning**.

One neuron

A single artificial neuron does three things:

1. Takes some inputs (numbers).
2. Multiplies each input by a **weight** (how important that input is), adds them up, and adds a **bias** (a starting offset).
3. Passes the total through an **activation function**, which decides how strongly the neuron "fires".

Example: a neuron deciding "should I carry an umbrella?" - Inputs: cloudiness = 0.9, rain forecast = 0.8, already raining = 0 - Weights: 0.5, 0.8, 1.0; bias: −0.6 - Total = 0.9×0.5 + 0.8×0.8 + 0×1.0 − 0.6 = 0.45 + 0.64 − 0.6 = **0.49** - A positive result → the neuron fires → "take the umbrella".

Learning means finding good values for all the weights and biases.

Layers

Layer	Job	Example: reading a handwritten digit
Input layer	Receives the raw data	784 numbers — one per pixel of a 28×28 image
Hidden layers	Build up patterns step by step	First edges, then curves and loops, then digit shapes
Output layer	Gives the answer	10 neurons, one for each digit 0–9; the highest wins

A network with **many hidden layers** is a **deep** network — that is where "deep learning" gets its name.

Why depth helps

Each layer builds on the one before. In image recognition, early layers detect simple **edges**, middle layers combine edges into **shapes** (eyes, wheels, leaf spots), and later layers recognise **whole objects**. This step-by-step building is why deep networks work so well on images, sound and language, where hand-written rules fail.

Activation functions

Without activation functions, stacking layers would still give only a straight-line model. Activations add the **curves** that let networks learn complex patterns.

- **ReLU** — keeps positive numbers, turns negatives into zero. Simple and very common.
- **Sigmoid** — squeezes any number into 0 to 1; useful for probabilities.
- **Softmax** — turns the output layer into probabilities that add up to 100% (for example, 92% "digit 7").

Special network designs

- **Convolutional neural networks (CNNs)** — designed for images; they scan small patches to find patterns anywhere in the picture. Used in crop-disease apps and medical scans.
- **Recurrent neural networks (RNNs)** — process sequences step by step; used for speech and text before transformers.
- **Transformers** — the modern design for language and more (Lesson 17).

The cost of power

Deep networks can have **millions or billions** of weights. They need lots of data, powerful GPUs and energy to train, and they are hard to explain — a challenge you will study in Week 4.

Remember this: A neural network is layers of neurons; each neuron weighs its inputs, adds a bias and applies an activation. Many layers let the network build simple patterns into complex ones.

Name: _____ Date: _____

Activity

Task 1

Redo the umbrella neuron with your own weights for a sunny day (cloudiness 0.2, rain forecast 0.1, already raining 0). Does it fire?

Task 2

Change one weight so the neuron becomes more "careful" (fires more easily). Explain your change.

Task 3

Describe in three sentences what the early, middle and late layers might detect in a network that recognises cassava diseases from leaf photos.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. What does a single artificial neuron do?

   A. Connects to Wi-Fi

   B. Weighs its inputs, adds a bias and applies an activation function

   C. Stores files

   D. Draws pictures

2. What is a weight in a neural network?

A. The output label
B. How heavy the computer is
C. The number of layers
D. A number showing how important an input is

3. What makes a network 'deep'?

A. It has no activation function
B. It has many hidden layers
C. It is stored underground
D. It uses only one neuron

4. In image recognition, what do early layers usually detect?

A. Names of people
B. Sounds
C. Simple edges
D. Whole objects

5. Why are activation functions needed?

A. They are not needed
B. Without them, stacked layers would only learn straight-line patterns
C. To delete data
D. To make the network colourful

6. Which network design is built especially for images?

A. Decision tree
B. k-means
C. Convolutional neural network (CNN)
D. Linear regression

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